

AWS Cloud Infrastructure Provisioning

Cloud & DevOps Internship - Task 2

Region: eu-north-1 (Stockholm)

Infrastructure Report

This report documents the infrastructure, security configuration, monitoring, evidence organization, and cost documentation for the completed Task 2 lab.

1. Project Overview

Task 2 moves the learner from local development into AWS cloud infrastructure. The assignment requires identity management, networking, compute, storage, monitoring, security, architecture documentation, and cost estimation. The final expected environment includes IAM configuration, a custom VPC, an Ubuntu EC2 instance, Nginx, S3, CloudWatch, and supporting documentation.

Project evidence for this repository is organized under `architecture/`, `screenshots/`, `cost/`, `deployment/`, and `evidence/`.

2. AWS Region

Region code: eu-north-1

Region name: Stockholm

The repository documentation uses this region consistently.

3. IAM Configuration

The Task 2 IAM work covers users, groups, policies, roles, and MFA/security configuration for the lab environment. The assignment also emphasizes using IAM rather than the root account for routine work and enabling MFA for privileged access.

- IAM users and user detail evidence are stored in `screenshots/iam/`.
- Group and access configuration evidence is stored in the IAM screenshot set.
- MFA/security configuration is represented in the IAM evidence.
- Sensitive credentials and private keys are excluded from the repository.

Evidence references: `screenshots/iam/iam-overview.png`, `iam-configuration.png`, `iam-users.jpeg`, `iam-devops.png`, `iam-cloud-intern.png`.

4. VPC and Networking

The networking design includes a custom VPC with public/private subnet configuration, an Internet Gateway, route tables, and Security Groups. These resources provide the network path required for the EC2 web-server workload while keeping administration and traffic rules explicitly documented.

- VPC configuration evidence: `screenshots/vpc/vpc-configuration.jpeg`
- Internet Gateway evidence: `screenshots/vpc/internet-gateway.jpeg`
- Route Table evidence: `screenshots/vpc/route-table.jpeg`
- Security Group evidence: `screenshots/security-groups/security-group.jpeg`

The exact CIDR ranges and resource identifiers are available in the corresponding AWS console screenshots and should be used as the source of truth for any future infrastructure changes.

5. EC2 Compute and Linux Server

An Ubuntu Linux EC2 instance was deployed as the compute resource for the lab. The instance was accessed using SSH, and Nginx was installed as the web server.

- EC2 configuration: `screenshots/ec2/ec2-configuration.jpeg`
- EC2 instances: `screenshots/ec2/ec2-instances.jpeg`
- SSH connection: `screenshots/ec2/ssh-connection.jpeg`
- Nginx verification: `screenshots/ec2/nginx-web-server.jpeg`

The assignment expects an Ubuntu EC2 instance, successful SSH access, and a working Nginx web server.

6. Security Groups and Access Control

Security Group rules were configured for the EC2 workload. The assignment specifically cautions against unnecessarily exposing SSH to 0.0.0.0/0 and emphasizes controlled access.

Repository evidence is available under `screenshots/security-groups/`.

7. Amazon S3 Storage

Amazon S3 was configured for object storage. The repository also contains evidence related to the static company website challenge.

- Bucket evidence: `screenshots/s3/buckets.jpeg`
- Configuration evidence: `screenshots/s3/s3-configuration.jpeg`
- Static site evidence: `screenshots/s3/s3-site.jpeg`
- Company static site evidence: `screenshots/s3/company-static-site.jpeg` and `s3-company-static-site.jpeg`

The assignment requires bucket creation, object upload/management, permission review, and versioning as applicable to the completed lab.

8. CloudWatch Monitoring

Amazon CloudWatch was configured for EC2 monitoring. The evidence set includes the service overview, a metrics graph, alarm views, and alarm configuration.

- CloudWatch overview: `screenshots/cloudwatch/cloudwatch-overview.jpeg`
- Metrics graph: `screenshots/cloudwatch/cloudwatch-graph.jpeg`
- Alarms list: `screenshots/cloudwatch/cloudwatch-alarms.jpeg`
- Alarm configuration: `screenshots/cloudwatch/cloudwatch-alarm.jpeg`

The task specifies CloudWatch metrics, CPU monitoring, instance health, and basic alerts.

9. Architecture Diagram

The repository includes the architecture diagram at `architecture/aws-architecture.jpeg`. This diagram is the visual representation of the deployed AWS components and their relationships.

The assignment requires an AWS architecture diagram as a final deliverable.

10. Cost Documentation

AWS Pricing Calculator evidence is stored under `cost/`. The repository contains `pricing-calculator.jpeg`, `pricing-estimate.jpeg`, `pricing-overview.jpeg`, and `pricing-payment.jpeg`.

Because the exact numeric estimate is visible only in the user-owned pricing screenshots and was not supplied as machine-readable data here, this report intentionally does not invent a dollar figure. The screenshots should be treated as the authoritative cost evidence.

11. Security and Repository Hygiene

- Do not commit AWS access keys, secret keys, passwords, MFA secrets, or private SSH keys.
- Use IAM identities for routine administration instead of the root account.
- Use MFA for privileged identities.

- Restrict SSH exposure to the intended source wherever possible.
- Stop or terminate unused lab resources to reduce unnecessary costs.

A .gitignore file is included in the repository to help prevent accidental inclusion of credentials and private keys.

12. Repository Evidence Map

The evidence-index.md file maps the main Task 2 requirements to the corresponding repository files. The deployment-documentation.md file provides a concise operational summary.

Primary evidence directories: screenshots/iam, screenshots/vpc, screenshots/ec2, screenshots/security-groups, screenshots/s3, screenshots/cloudwatch.

Primary documentation: README.md, deployment/deployment-documentation.md, evidence/evidence-index.md.

13. Deliverable Checklist

Deliverable	Repository location	Status
AWS Architecture Diagram	architecture/aws-architecture.jpeg	Provided
GitHub Repository	Repository root	Provided
Infrastructure Report	docs/infrastructure-report.pdf	This report
IAM Evidence	screenshots/iam/	Provided
VPC Evidence	screenshots/vpc/	Provided
EC2 Evidence	screenshots/ec2/	Provided
Security Group Evidence	screenshots/security-groups/	Provided
S3 Evidence	screenshots/s3/	Provided
CloudWatch Evidence	screenshots/cloudwatch/	Provided
README	README.md	Provided
Cost Estimation Evidence	cost/	Provided
Deployment Documentation	deployment/	Provided

The assignment also permits enhancement ideas such as bastion hosts, separate environments, user-data automation, dashboards, budgets, CLI scripts, and detailed diagrams. These are optional enhancements and are not required to claim completion of the base Task 2 deliverables.

14. Conclusion

The repository contains the core evidence and documentation for the AWS Cloud Infrastructure Provisioning task: IAM, VPC networking, EC2, SSH, Nginx, S3, CloudWatch, architecture, security documentation, cost evidence, and deployment documentation. The remaining project-level action is to keep the repository synchronized with any final report edits and to remove unused AWS resources when the lab is no longer needed.